

Yuqian Zhang

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EDUCATION

Duke Kunshan University (DKU) & Duke University Dual Degree 08/2022-05/2026
B.S. in Applied Math & Computational Science; Math Track (DKU) Kunshan, China
B.S. in Interdisciplinary Studies: Applied Math & Computational Science; Math (Duke) Durham, U.S.

- **GPA: 3.85/4.0**; Dean's List Fall 2023, Spring 2024, Spring-Fall 2025 (**w. Distinction**)
- **Relevant Coursework:** Stochastic Differential Equation, Measure Theory, Numerical Analysis, Advance Linear Algebra, Algorithm Analysis & Design, Graphics Software Architecture, Image Data Science.

RESEARCH EXPERIENCE

Research Intern with Dr. Yi Zhang, Professor of Neurosurgery 12/2024-Now
Duke University, Lab: [Yi Zhang CompBio @ Duke](#) | [yi-zhang-compbio-lab](#) Durham, U.S.

Project 1. PinFuse: Spatial Transcriptomics Data Denoising through Neural Advection-Diffusion Operator

- Developed **PinFuse**, a lightweight Graph VAE with Position-induced Transformer (PiT) variants for denoising spatial transcriptomics data. Proposed a neural advection-diffusion operator that preserves mass conservation while modeling zero-inflated negative binomial observations.
- Incorporated Pyro for probabilistic inference and achieved improved gene reconstruction, latent clustering, and gene purity over recent methods including ResolVI.

Project 2. Gene Expression Field Estimation via Poisson Resampling and Neural ODEs

- Formulated spatial transcriptomics data as a spatial Poisson sampling process over a diffused gene expression field and developed a reversed Neural ODE framework for field reconstruction from sparse observations, combining supervised pretraining with semi-supervised operator learning.

Research Assistant with [Dr. Zuchuan Li](#), Professor of Data and Environmental Sciences 11/2023-Now
Duke Kunshan University Kunshan, China

Project 1. MissingJiT: Learning the Joint Distribution of Satellite Imagery from Missing Data

- Developed a flow-matching framework for learning the joint distribution of incomplete satellite imagery and performing posterior sampling for image inpainting.
- Designed a pixel-space JiT with missing-patch tokenization and masked attention, and derived a flow-matching analogue of Diffusion Posterior Sampling (DPS) with an SSIM-based likelihood for preserving fine-scale structure.

Project 2. Solving Time Series Inverse Problem by Combining Bayesian-PINN with Kalman Filter

- Proposed a Bayesian PINN-Kalman Filter framework for time-series inverse problems, including a patched covariance method to improve filtering performance; implemented a CUDA-accelerated differentiable 2D advection-diffusion solver as a neural operator.
- Conducted an independent study of score-based diffusion models for inverse problems, deriving and implementing SDE-based diffusion processes and Diffusion Posterior Sampling (DPS); reproduced DDPM and other sampling methods for both linear and nonlinear inverse problems.

WORKING PAPER

Zhang, Y., Shen, Z., Deng, Y. and Zhang, Y. "PinFuse: Spatial Transcriptomics Data Denoising through Neural Advection-Diffusion Operator" Manuscript, 2026.

Zhang, Y. and Li, Z.. "MissingJiT: Learning the Joint Distribution of Satellite Imagery from Missing Data" Manuscript, 2026.

PROFESSIONAL EXPERIENCES

Baidu, Inc., Famo Agent 07/2026-10/2026
Evolving Algorithms & AutoResearch System Development Beijing, China

- Developed an evolving algorithm that combines zero-order optimization with method-space embeddings to perform pseudo-gradient descent over mid-scale code spaces.
- Worked an autonomous research agent for biological discovery, focusing on skill routing, skill distillation, and evolutionary improvement. Surveyed agentic systems including AutoResearchClaw and Biomni; reproduced

experiments on BixBench and GeneBench-Pro; and distilled protein-design skills from open-source skill pools with ProteinGym Bench for the Famou-Scientist project.

SELECTED TECHNICAL PROJECTS

WebGL Implementation of Reflective Gaussian Splatting (Ref-Gaussian) GitHub <i>Signature Work</i>	08/2025-03/2026 Kunshan, China
- Implemented the Ref-Gaussian pipeline in WebGL, achieving 60+ FPS real-time rendering with optimized loading and memory efficiency. Proposed Approximate Tangent Axis (ATA) estimation for Gaussian Surfels rasterization and integrated culling and lazy-update strategies to improve rendering efficiency. - Built an end-to-end pipeline from COLMAP reconstruction and Ref-Gaussian inference to Docker-based web deployment.	
Blender Development Blender Projects <i>Individual Contributor</i>	02/2024-Now Remote
Enhanced Blender’s curve sculpting and geometry editing pipeline, implementing curve radius initialization, interpolation, UI integration, and related documentation; also improved several material property editing and curve-mapping interactions. 5 pull requests accepted into the official Blender repository.	
CUDA IP Operator Development GitHub <i>Independent Project</i>	9/2024-12/2024 Kunshan, China
Developed a CUDA-accelerated Python library for Image Processing (IP) and Computer Vision, achieving high-throughput GPU performance and scalable execution. Implemented core algorithms including convolution, geometric transformations, dithering, morphological operations, and texture segmentation.	
GPU(OpenGL) based Renderer Development GitHub <i>Independent Project</i>	05/2022-06/2026 China
- Built a real-time rendering engine with 30,000+ lines of C++/OpenGL code across 600+ commits. Studied 30+ foundational and state-of-the-art graphics papers, including reproductions of 3 SIGGRAPH works. - Conducted in-depth exploration of physically based rendering, high-quality shadowing, signed distance fields, spatio-temporal denoising, and real-time global illumination.	

TEACHING

CS101 Teaching Assistant	Fall 2025 Kunshan, China
Led weekly office hours and lab sessions for programming fundamentals, debugging, and problem-solving, supporting dozens of students with 12 hours of instruction per month	
MATH105 Teaching Assistant	Fall 2025 Kunshan, China
Led weekly office hours and recitation sessions, reinforcing mathematical foundations, conceptual understanding, and rigorous problem-solving skills.	

AWARDS & SCHOLARSHIPS

Magna Cum Laude - Latin Honors	2026
DKU Student Experiential Learning Fellowship Program	Summer 2025
Duke Undergraduate Semester Research Program Reward	Spring 2025
DKU Summer Research Scholars	Summer 2024
Mathematical Contest in Modeling 2024 C - Honorable Mention	Spring 2024
MathorCup 2023 – Big Data Competition - Second Award	Fall 2023

SKILLS

Programming: C++, Python (PyTorch), CUDA, Mathematica.
Computer Graphics: OpenGL, WebGL, GLSL, COLMAP
Tools: Git, Docker, Blender, PhotoShop